



C23-CM-CCB-AI-CAI-304

23191

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME – THIRD SEMESTER EXAMINATION

DATA STRUCTURES THROUGH ‘C’

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Define sorting and write the need for sorting.
- 2. Write about nonlinear data structure and give examples.
- 3. Write the advantages of linked lists over arrays.
- 4. Write the purpose of dummy header in a linked list.
- 5. List any three applications of stacks.
- 6. If $p = 4$, $q = 14$ and $r = 9$ then evaluate the postfix expression $pqr+ \quad /$ and write the result.
- 7. Write about Queue and its operations.
- 8. What is the main drawback of Queue and how it can be resolved?
- 9. Define binary tree.
- 10. Write the differences between Binary Tree and General Tree.

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[Contd...

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** Explain binary search method with an example and write a C program to implement recursive version of binary search.
- 12.** Write a 'C' program to create and display a doubly linked list.
- 13.** Explain the array implementation of stack and its operations with examples.
- 14.** (a) Write a 'C' program to convert an infix expression into a postfix expression. 5
(b) Convert the infix expression $(A-B)*C+D/E$ into postfix notation using stack. 5
- 15.** Write a 'C' program to implement circular Queue and its operations using linked lists.
- 16.** (a) Explain priority Queues. 5
(b) Write the applications of Queues. 5
- 17.** Explain the construction of a binary tree for the given inorder and post order traversals.
Inorder : D G B A H E I C F Postorder : G D B H I E F C A
- 18.** Explain various binary tree traversal methods with algorithms and examples.

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